Sumangalam Avtar Sharma

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EDUCATION

Indian Institute of Science Education and Research, Pune, Maharashtra, India Aug. 2024 – Present

- **BSMS Integrated Dual Degree Program**

ITM Global School, Gwalior, Madhya Pradesh, India Apr 2019 - Nov 2023

- **Cambridge AS and A Levels**
- **Cambridge IGCSE**

ACADEMIC PROJECTS

Parameterization of Lorenz96 and Testing Online Stability Against Training Data Volume Aug 2026 - Present

Research Project Independent

- Replicated M^2LInES notebooks on Lorenz96 model
- Current RQ: What minimal training-data volume is required for a neural network parameterization of the Lorenz-96 subgrid coupling term U_k to remain stable and physically consistent when coupled online into the reduced (slow-variable only) model as opposed to merely being accurate offline?
- Conducted a data-volume sweep over 10 different data volumes (10 total number of steps) for 2 different types of models; each activated by a different activation function
- Trained and evaluated 20 different models based on different metrics learnt from M^2LInES to check for long-horizon stability online

Patterns of Reasoning and Factors of Growth Jun 2026 - July 2026

Research Project ProblemSolvers Lab | Department of Science Education, IISER Pune

* Conducted an extensive literature review scoping out work done in the past on the topic
* Carried out multiple rounds of data cleaning and organization
* Created and synthesized a new conceptual framework to understand the reasoning patterns in pattern recognition problems
* Documented a pre-existing rubric used a measure of performance

Impact of ENSO on Rainfall Variability Across Major Kharif Crop Growing States May 2026 - July 2026

Research Project and Training Program IDEAS-TIH, ISI Kolkata

* Developed an understanding on various topics during the course of the training program including but not limited to Basic Statistics for Data Science and Data Visualization, GitHub and Cloud Computing, Streamlit, ML and LLM Fundamentals, Deep Learning, Computer Vision, Agentic AI and Reinforcement Leaning
* Carried out EDA for the provided rainfall data
* Developed a interpolation method to obtain a higher resolution rainfall data as compared to the one obtained by IMD Portal of India
* Used Google API and Cloud Project to automate the Google Sheets workflow using Python Scripts to make the process modular for each state
* Developed a metric to quantify rainfall variability
* Classified rainfall into various categories by the help of various metrics including ENSO Sensitivity Index, Classification Balance etc.

Metric Spaces Dec 2025 - Feb 2026

Reading Project IISER Pune

* Read through the copy of "Metric Spaces" by "Micheal O Searcoid"
* Solved problems spanning all sets of difficulty on the topic, including but not limited to problems related to the various coding metrics such as Hamming and Levenshtein metrics, p-adic metric, linear normed spaces etc.

Statistical Instrument on Circular Economy Jun 2025 - Oct 2025

REYES Mentorship Program UC Berkeley

* Conducted a survey targeting over 1000 responses to better understand the awareness about Circular Economy in the general masses
* Collaborated on a mass literature review spanning over 300 articles to assess the current research scenario on Circular Economy
* Analyzed research questions targeting qualitative and quantitative aspects of the data extracted from the literature review
* Performed in-depth analysis of survey data to present findings in a formal manner by techniques such as hypothesis testing, ANOVA and regression analysis

LEADERSHIP ROLES

Coordinator Jul 2025 - Present

The Maths Club IISER Pune

* Designed a discord bot to host the annual Treasure Hunt event
* Hosted Pi-Week celebrations
* Managed various departments of the club, including but not limited to Publicity, Media, Outreach, Logistics etc.
* Organized various talks spanning a vast domain of Mathematics
* Designed questions for events, including advanced topics such as Probability Theory, Surface Integrals, Metric Spaces and Cryptography
* Conducted multiple publicity drives in-person and on Instagram
* Hosted outreach drive to publicize club activities

Social Media and Logistics Coordinator Sep 2025 - May 2026

Aroha, The Music Club IISER Pune

Sponsorship Executive Oct 2024 - May 2025

Entrepreneurship and Innovation Cell IISER Pune

* Spearheaded multiple sponsorship drives
* Managed corporate relationships and ensured all deliverables, such as logo placements and event stalls, were met
* Attended and established connections in multiple E-summits

Event Coordinator Sep 2024 - May 2025

Aroha, The Music Club IISER Pune

VOLUNTEERING ROLES

Question Making Team Nov 2024 - Mar 2025

Mimamsa IISER Pune

* Designed questions spanning multiple fields of Physics, from material properties of solids to mechanical feasibility of an apparent perpetual machine

SciCon (Science Convention) Volunteer Mar 2025

Mimamsa IISER Pune

* Led a poster presentation on the fundamentals of working of electric guitars, explaining in great detail the mathematical and physical laws binding it

Quizzing Volunteer Mar 2025

Mimamsa IISER Pune

SKILLS

Languages : Python, R, HTML/CSS, Markdown, LaTeX, Bash

Tools : Matplotlib, Desmos, Geogebra, NumPy, Scikit, SciPy, AutoCAD, DaVinci Resolve, Canva, Streamlit